A critical evaluation of the Gompertz model in pmsimstats-ng

2026-05-07 06:11 PDT

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Purpose. This note evaluates the modified Gompertz function that pmsimstats-ng uses as the mean trajectory of its three latent response components. It sets out where the form is defensible, which assumptions it imposes without defense, how those assumptions bear on the biomarker-treatment interaction test, and which alternative families could serve as sensitivity analyses. The argument is from the literature and from inspection of the package source; no simulation was run for this note. Several of its recommendations were later taken up by paper 07 (`analysis/report/07-gompertz-evaluation/`), and dated notes in Sections 6 to 8 point to that work.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here. Code identifiers appear in backticks.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
t	time in weeks (the Gompertz argument)	canonical
$y(t)$	value of a response-curve family at time t	local
a, g_b, g_c	asymptote, displacement and rate of the textbook Gompertz form (code `maxr`, `disp`, `rate`)	local
ϵ_{mach}, expm1(x)	double-precision machine epsilon; $e^x - 1$ evaluated without cancellation	local
$E_{\max}$, EC_{50}, h	maximum effect, half-maximal time and Hill exponent of the Emax/Hill family	local
BR_{it}, PB_{it}, TV_{it}	biological, placebo-belief and time-variant response components	canonical
Y_{it}	observed symptom score (code `Sx`)	canonical
B_i, b_i	biomarker and standardized biomarker (code `bm`, `bm_z`)	canonical
$D_{bc,it}$	exposure-decayed drug indicator (code `Dbc`)	canonical
$\beta_{bm:D}$	biomarker-by-treatment interaction coefficient (`bm:Dbc`)	canonical
β_D	treatment main effect	canonical
β_0, β_B, β_t	intercept, biomarker main effect and linear time slope of the analysis model	local

Symbol	Meaning	Status
u_i, ε_{it}	participant random intercept; residual	canonical
c_{bm}, β_{bm}	covariance-moderation correlation; mean-moderation multiplier	canonical
σ_{BR}	standard deviation of the BR component	canonical
$t_{1/2}$, t_{od}	carryover half-life; time on drug (code `tod`)	canonical
N	total sample size across randomization paths	canonical
k	Weibull shape parameter	canonical
λ_Z, μ_m	lag time and maximum specific growth rate of Zwietering's modified Gompertz	local
a_i, a_0, τ_a^2	participant-level asymptote, its mean and variance (Recommendation 5)	local

- **Modified Gompertz (pmsimstats sense).** The Gompertz curve shifted so that $y(0) = 0$ and rescaled so that $y(\infty) = a$ (code `modgompertz`).
- **Covariance-moderation architecture.** The interaction lives in the on-drug B_i-BR correlation c_{bm} (formerly Architecture B).
- **Mean-moderation architecture.** The interaction is an on-drug shift $\beta_{bm} b_i \sigma_{BR}$ of the BR mean (formerly Architecture A).
- **Inflection.** The time at which the curve's slope is greatest.

1. Motivation and scope

The pmsimstats-ng simulation framework uses a modified Gompertz function as the parametric template for three latent response components: the biological response (BR), the time-varying non-drug component (TV), and the placebo-belief component (PB). The function appears in `R/utilities.R` and in `implementations/{original,original-extended,tidyverse}/R/` collections; it is invoked by the simulation core at every cell of every parameter grid. Because the assumed functional form of the response trajectory is loaded into the simulated outcome before any analysis-model question is asked, the choice of family is not a peripheral implementation detail but a foundational

modeling decision. This document evaluates that decision against the relevant biological and biostatistical literature, articulates where the Gompertz form is well-justified for the present setting, identifies where it imposes assumptions the framework does not explicitly defend, and outlines the alternative families that could be used in sensitivity analyses.

2. The Gompertz function

The Gompertz function in its original form is a double-exponential sigmoid

$$y(t) = a \cdot \exp(-g_b \cdot \exp(-g_c \cdot t)),$$

where $a > 0$ is an asymptote, $g_b > 0$ controls displacement of the inflection along the time axis, and $g_c > 0$ is a rate parameter [1]. The function was introduced as a demographic model of human mortality and was first re-purposed as a growth curve by Winsor [2], who observed that the sigmoidal shape of biological size-by-time data could be captured parsimoniously by the same three-parameter form. Subsequent work established the Gompertz model as a standard description of tumor growth [3, 4, 5], bacterial population dynamics [6], and innovation diffusion. A review by Tjørve and Tjørve [7] catalogs parameterizations and emphasizes the non-orthogonality of g_b and g_c, which complicates identification when both are estimated from data.

In pharmacodynamic modeling, the dominant family is the Hill or Emax model, $y = E_{\max} \cdot t^h / (EC_{50}^h + t^h)$ [8, 9, 10], which differs from the Gompertz in two respects relevant here. First, the Hill model is symmetric about its half-maximal point on the logarithmic time scale (it is a logistic function of $\ln t$); on the linear time scale it is not symmetric, and for $h > 1$ its inflection lies at $EC_{50}\,[(h-1)/(h+1)]^{1/h}$, before EC_{50} (derived). Second, its half-maximal point EC_{50} has a direct pharmacological interpretation (half-maximal effect concentration, or time when the argument is time). The Gompertz inflection occurs at $t = \ln(g_b)/g_c$ and has no comparable interpretive anchor in the drug-response literature.

3. The modified form used in pmsimstats-ng

`R/utilities.R` implements

```
y <- maxr * exp(-disp * exp(-rate * t))
vert_offset <- maxr * exp(-disp * exp(-rate * 0))
y <- y - vert_offset
y <- y * maxr / (maxr - vert_offset)
```

This is the Gompertz function shifted vertically so that $y(0) = 0$ and rescaled so that $y(\infty) \to$ `maxr`. The shift plus rescale maps the family { Gompertz with arbitrary $y(0)$ and asymptote `maxr` } onto the family { Origin-passing curves with asymptote `maxr` }. The adjustment is convenient for symptom-improvement modeling because clinical outcome scales (CAPS, PCL, HAM-D, BDI) are typically expressed as change-from-baseline so that 0 is the correct value at study entry.

Two cautions about this implementation. First, the rescaling factor `maxr / (maxr - vert_offset)`, which equals $1/(1 - e^{-g_b})$ because `vert_offset` is $a\,e^{-g_b}$, grows as $1/g_b$ as `disp` shrinks because `vert_offset` approaches `maxr`, and the evaluation loses accuracy there (derived). (Corrected 2026-10-08: this sentence previously located the instability in the rescaling factor itself. The factor is benign in exact arithmetic, and the curve has the finite limit $a(1 - e^{-g_c t})$ as $g_b \to 0$. The loss of accuracy is floating-point cancellation in the numerator $\exp(-g_b e^{-g_c t}) - e^{-g_b}$, which subtracts two numbers close to 1; the relative error is of order $\epsilon_{\text{mach}}/g_b$, about 6×10^{-9} at $g_b = 10^{-8}$ and 3×10^{-4} at $g_b = 10^{-14}$ (at $g_c = 0.42$, $t = 5$), and $g_b = 0$ returns `NaN`. It is negligible for $g_b \gtrsim 10^{-6}$. Writing the curve as $a\,[\text{expm1}(-g_b e^{-g_c t}) - \text{expm1}(-g_b)]/[-\text{expm1}(-g_b)]$ is algebraically identical and accurate to machine precision (verified, `analysis/scripts/compendium-mathematics/check-gompertz-rmanova-cycles.R`). See `analysis/report/compendium-mathematics.tex`, Section "Shipped parameters, landmarks, the displacement, and numerical conditioning" (`sec:invc-gomp-landmarks`).) The current code guards against the degenerate `maxr == 0` case but not against a degenerate `disp` close to zero (inspected, `R/utilities.R`, 2026-10-07). Second, the term 'modified Gompertz' in the microbiology literature [6] denotes a different reparameterization, in which a lag time λ_Z and a maximum specific growth rate μ_m are explicit parameters; the pmsimstats-ng usage is unrelated to that established modification, and overloading the name in code comments and manuscripts invites confusion.

4. How Hendrickson et al. (2020) use it

Hendrickson et al. [15] use the modified Gompertz to encode three components of the simulated symptom trajectory. The biological response (BR) is a function of time on drug t_{od}; the time-varying non-drug component (TV) is a function of calendar time since study entry; the placebo-belief component (PB) is a function of cumulative time under positive expectancy, scaled by the expectancy weight. Each component receives its own (maxr, disp, rate) triple. (In all current pmsimstats-ng DGPs t_{od} restarts at 0 on re-exposure; it is time since the current exposure began, not lifetime cumulative exposure.) The published calibration, the package dataset `extracted_rp`, is

Table 2: Published (Hendrickson) modified Gompertz parameters for the BR, TV, and PB components, from the package dataset `extracted_rp`

Component	maxr	disp	rate (per week)	sd
BR	10.99	5	0.42	8
TV	6.51	5	0.35	10
PB	6.51	5	0.35	10

The carryover-sensitivity driver (`default_resp_param()` in `analysis/scripts/carryover-sensitivity/sim` departs from this: it gives TV the BR curve (10.99, 5, 0.42) and uses standard deviations of 6, 2 and 8 for TV, PB and BR. Both sets were read from the source on 2026-10-07 (inspected); an earlier version of this table listed the driver values as the Hendrickson defaults.

The published curves rise from zero at study entry and pass through inflection near $t = \ln(5)/0.42 \approx 3.8$ weeks for BR and $\ln(5)/0.35 \approx 4.6$ weeks for TV and PB. At week 12 the BR curve has reached about 97 percent of its asymptote (derived). The calibration target is the prazosin-versus-placebo CAPS trajectory of the Raskind trials [16, 17, 18]. The BR-versus-TV-versus-PB separation in the Hendrickson model is an architectural choice rather than a mechanism testable from the published data: the Raskind trials report the total CAPS difference, not a decomposition into biological-response and non-drug components. The package documentation of `extracted_rp` says as much: it calls the set a 'tabula rasa' in which the time course and maximum of the expectancy (PB) and time-varying (TV) factors were assumed equal. This identifiability constraint is acknowledged in the package documentation but is not, to our knowledge, evaluated empirically in the published literature.

5. Strengths in the symptom-trajectory setting

Sigmoidal shape. Drug-induced symptom improvement in chronic neurobehavioral conditions is widely observed to be sigmoidal rather than linear: there is a delay before measurable benefit, followed by a phase of more rapid improvement, followed by a plateau as the underlying pathophysiology stabilizes [10]. A monotone three-parameter sigmoid is therefore a defensible parametric template for the mean structure of such trajectories, and the Gompertz is one of several families that can produce this shape.

Parsimony. Three parameters per component is the minimum needed to represent magnitude, timing, and rate, which is attractive for a simulation framework that must enumerate many parameter grids without combinatorial explosion. Richer families (four-parameter logistic, Weibull cumulative, splines) introduce identifiability problems when the simulation is calibrated to a single observed trajectory.

Computational tractability. The Gompertz is closed-form, infinitely differentiable, and free of the behavior of the Hill function at $t = 0$, where the slope is unbounded for $h < 1$ and zero for $h > 1$ (derived). The double-exponential form is numerically stable across the parameter ranges used in the framework.

Established precedent in adjacent fields. The Gompertz has been used for decades in tumor-growth modeling [3, 4, 5] and bacterial-growth modeling [6], giving the family a level of biological face validity that less commonly used sigmoids do not enjoy.

6. Limitations and concerns

Asymmetry that may not match the data. The Gompertz is more strongly skewed than its biological cousins (logistic, Hill, Weibull cumulative). Its inflection occurs at $1/e$ of the asymptote (approximately 37 percent), rather than at 50 percent as in the logistic; for the origin-shifted pmsimstats form with $g_b = 5$ the fraction is $(e^{-1} - e^{-5})/(1 - e^{-5}) \approx 0.36$ (derived). Whether prazosin-induced CAPS improvement in PTSD truly accelerates earlier than it decelerates is an empirical question; the published Raskind trajectories [19] are not granular enough to settle the question of skewness, and the Hendrickson framework does not report sensitivity to the choice of sigmoid family. (*Note, 2026-10-07:* paper 07 has since run that sensitivity; see the note in Section 6b.)

Monotonicity. The Gompertz cannot accommodate response followed by relapse, which is documented in PTSD pharmacotherapy [20, 21] and chronic depression. Patients who improve early in treatment and then lose benefit at week 8-10 cannot be simulated under any $(\text{maxr}, \text{disp}, \text{rate})$ choice; the curve is monotone non-decreasing by construction. For a simulation framework whose downstream analyses include the OL+BDC design (open-label titration followed by blinded discontinuation), the inability to simulate within-treatment relapse is a meaningful restriction. The pmsimstats-ng framework currently models relapse only as off-drug carryover decay, which conflates two distinct biological mechanisms (treatment escape during continued dosing, versus pharmacological washout after discontinuation).

Plateau is fixed. The asymptote `maxr`, like `disp` and `rate`, is a population-level parameter; there is no participant-level random effect on any Gompertz parameter. Participant heterogeneity enters instead as additive deviations around the common mean curve: each component is drawn from a multivariate normal with a constant per-timepoint standard deviation (`respparam$sd`) and AR(1) serial correlation, and under the mean-moderation architecture the on-drug BR mean is shifted by $\beta_{bm}\, b_i\, \sigma_{BR}$ (inspected, `buildSigma()` and `generateData()` in `R/generateData.R`, 2026-10-07). Heterogeneity of treatment effect, the phenomenon that motivates predictive-biomarker work, is therefore grafted on at the architecture layer (the covariance-moderation correlation c_{bm} or the mean-moderation shift), not as a property of the Gompertz family. Every participant

shares the same curve shape and timing; individuals differ in level, not in rate or ceiling. Under the default `moderation_scaling = 'constant'` the biomarker shift has the same size early in exposure, when the mean BR is near zero, as at the plateau; the `'trajectory'` option scales it with the mean BR curve instead. That a single shape with level-only heterogeneity fits individual prazosin responders is a strong assumption that is not, to our knowledge, defended empirically. (*Correction, 2026-10-07:* an earlier version said participants approach a common ceiling at different rates; the source shows no rate heterogeneity.)

Non-orthogonal parameters. The displacement parameter g_b (`disp`) and the rate parameter g_c (`rate`) are strongly correlated in the Gompertz likelihood surface [7]. In a calibration setting this is a nuisance; in a simulation setting this manifests as parameter-grid axes that are not really independent. A grid that varies `disp` while holding `rate` fixed is also implicitly varying the inflection time and the early-trajectory slope, which may be undesirable for sensitivity analysis. The pmsimstats-ng parameter-sensitivity sweeps in `analysis/scripts/carryover-sensitivity/` and `analysis/scripts/nof1-design-sensitivity/` do not currently isolate these geometric features of the Gompertz curve.

Vertical-offset rescaling can amplify noise. The $y - y(0)$ shift plus rescale produces a curve whose value at any given t is sensitive to small changes in `disp` because both `vert_offset` and the rescale factor depend on it. The function is well-behaved in the parameter regime used by Hendrickson et al. [15] but the framework does not currently flag the regions where the rescale factor diverges.

Departure from established pharmacodynamic practice. The dominant pharmacodynamic family is the Emax/Hill model [8, 9, 10]. Reviewers familiar with the PK-PD tradition may ask why a Gompertz was preferred to an Emax model whose half-maximal-time parameter has a direct interpretation. The pmsimstats-ng default calibration does not currently report a sensitivity analysis under an Emax alternative, and the published Hendrickson paper does not defend the Gompertz choice over a Hill or Emax form on biological grounds.

Misspecification relative to the analysis model. The analysis model in pmsimstats-ng is a linear mixed-effects model (`nlme::lme(Sx ~ bm + t + Dbc + bm:Dbc, ...)`) with a continuous exposure-decayed predictor $D_{bc,it}$ (`Dbc`). Under the Gompertz DGP, the true mean structure is a sum of three sigmoidal trajectories; the analysis fits this with a linear-in-time term plus a binary-or-decayed-exposure term. The mismatch is intentional, and the simulation studies the consequences for power and Type I error, but the mismatch does mean that the simulated power estimates are conditioned on a specific functional form of the DGP that is itself a modeling choice rather than an established fact.

6b. Implications for the biomarker-treatment interaction test

The founding pmsimstats question is the test of the biomarker-treatment interaction in the mixed-effects model `Sx ~ bm + t + Dbc + bm:Dbc` with a random intercept per participant, that is $Y_{it} = \beta_0 + \beta_B B_i + \beta_t t + \beta_D D_{bc,it} + \beta_{bm:D} B_i D_{bc,it} + u_i + \varepsilon_{it}$, with $H_0 : \beta_{bm:D} = 0$ [15]. The Gompertz-induced response trajectory enters the simulated data through the BR component mean. The limitations enumerated in Section 6 do not contribute equally to the validity of this interaction test. Ranking them by their impact on the bias, power, and coverage of $\widehat{\beta}_{bm:D}$ (argued, not simulated):

Most critical for the interaction test.

1. **Common shape, level-only heterogeneity (no random effect on `maxr`).** The covariance-moderation architecture encodes the interaction signal in the second-moment B_i-BR correlation; the mean-moderation architecture encodes it in a first-moment mean shift. In both, between-participant heterogeneity in the biomarker-driven treatment effect is a difference in *level* around a common curve: no participant reaches a different ceiling through a different `maxr`, and none responds faster or slower (Section 6). The biomarker-stratification literature typically conceptualizes responder-versus-non-responder heterogeneity as different asymptotic benefit [23, 24]. A level shift of constant size is close to that at the plateau but not early in exposure, so the restriction matters most for designs whose on-drug observations fall on the rising part of the curve. The direction of any resulting bias in simulated power is not established; a plateau-heterogeneous DGP would be needed to measure it. A reviewer who accepts the latent-subtype framing of the covariance-moderation architecture will reasonably ask why the response ceiling is treated as a population constant.

2. **Monotonicity (no within-treatment relapse).** The Gompertz cannot generate response-then-relapse trajectories. In real PTSD pharmacotherapy data [20, 21] a non-trivial fraction of participants show early benefit followed by decay, often correlated with comorbid depression and chronic pain status. If the true symptom trajectory has this shape and the simulation does not, the linear-in-time analysis model is being asked to fit a smoother surface than it would encounter in practice. The interaction test is then evaluated under conditions kinder than the application; this plausibly biases the simulated power upward (optimistic). Because the direction of the point (1) effect is not established, the two cannot be assumed to offset.

3. **Misspecification relative to the linear analysis model.** The analysis model fits `bm + t + Dbc + bm:Dbc`. The true BR mean is sigmoidal, not linear in t. The linear-in-t term absorbs trajectory curvature into a single

slope, and any residual curvature is pushed into the random intercept and the AR(1) residual structure. The interaction term `bm:Dbc` is the residual quantity that survives this absorption; it is therefore the term most exposed to the choice of sigmoid family. A Hill-shaped DGP (more symmetric inflection) and a Gompertz-shaped DGP (asymmetric inflection at 1/e of the asymptote) push different amounts of curvature into the linear approximation and may yield different apparent `bm:Dbc` standard errors at matched calibration targets. The framework currently does not report sensitivity under an alternative sigmoid family; this is the single most actionable Gompertz-related contribution to interaction-test validity.

Note, 2026-10-07. Paper 07 (`analysis/report/07-gompertz-evaluation/`) has since run this sensitivity: a 16-cell factorial (four families, both architectures, two effect sizes; $N = 70$, OL+BDC, $t_{1/2} = 0$, 1,000 replicates per cell). As reported in its abstract (inspected, not re-run here), the family is exactly inert under covariance moderation, where matched seeds give identical rejection decisions; under mean moderation with trajectory-scaled moderation the four powers span 0.743 to 0.782, with the Gompertz lowest. The 144-cell extension with carryover remains open.

Moderate impact on the interaction test.

4. **Asymmetric inflection.** The Gompertz inflection at $t = \ln(g_b)/g_c$ corresponds to $1/e \approx 0.37$ of the asymptote, in contrast to the logistic 0.5. Under matched `maxr` and matched inflection time, the Gompertz therefore places less of its rise before the inflection and more after it than the logistic does. Under the mean-moderation architecture with trajectory-scaled moderation, the interaction signal follows the BR curve, so the family changes the mix of contributions across timepoints and the absolute power numbers, though plausibly not the qualitative conclusions (argued). Under the covariance-moderation architecture the interaction lives in the covariance and is mean-independent, so the family does not affect it; paper 07 confirms this (see the note under point 3). (*Correction, 2026-10-07:* an earlier version of this point said the covariance-moderation signal was weighted toward the early-trajectory observations; paper 07 shows it is not.)

5. **Non-orthogonal `disp` and `rate`.** The displacement and rate parameters are correlated in the Gompertz likelihood surface [7]. For interaction-test simulation, this matters when sensitivity sweeps vary one of these parameters. A sweep that walks `rate` while holding `disp` fixed implicitly co-varies the inflection time and the early slope; the resulting effect on the interaction power is a mixture of two geometric properties of the curve, not a clean single-axis sensitivity. The carryover-sensitivity and nof1-design-sensitivity papers do not currently decompose this. A cleaner reparameterization (Tjørve's alternative form, or fixing the inflection time as the

primary parameter and deriving `disp` as a function of it) would produce sensitivity sweeps with clearer interpretation.

Low impact on the interaction test.

6. **Vertical-offset rescaling instability.** A numerical concern at extreme parameter values. The default ($\text{disp} = 5, \text{rate} = 0.42$) regime is far from the instability region; the interaction test is not affected at the published parameter settings. A guard in the code is prudent but the issue does not undermine current results. (Corrected 2026-10-08: the instability is cancellation in the numerator at very small `disp`, not the rescaling factor; an `expm1` form removes it. See Section 3.)
7. **Naming overlap with Zwietering's modified Gompertz.** A documentation concern, not a statistical concern. The interaction test is unaffected.
8. **Departure from Emax/Hill pharmacodynamic tradition.** A reviewer-facing concern about defensibility of the choice. The interaction test is unaffected at fixed parameters; it would change under an Emax DGP only through the same mechanism as point (3) above.

Summary. The Gompertz-related issues that most warrant attention before the next interaction-test manuscript revision are the fixed plateau (point 1), the monotonicity assumption (point 2), and the absence of a sensitivity sweep over sigmoid family (point 3). Points 4 and 5 are second-order. Points 6 through 8 are housekeeping. A direct sensitivity sweep at fixed ($\text{maxr}, \text{inflection time}$) across Gompertz, logistic 4PL, and Hill sigmoids would address point 3 and is the highest- value next experiment for the founding interaction-test question. (*Note, 2026-10-07:* paper 07 ran a version of it with the logistic, hyperbolic-tangent and piecewise-linear families; the Hill family was not included.)

7. Alternative parametric families to consider

Table 3: Alternative parametric response families with strengths, weaknesses, and references

Family	Strength	Weakness	Reference
Logistic (4PL)	Symmetric inflection at 50% asymptote	Same monotonicity problem as Gompertz	[11]

Family	Strength	Weakness	Reference
Hill / Emax	Closed-form pharmacological interpretation; standard in PK-PD	Unbounded slope at $t = 0$ when $h < 1$; less natural for time courses than for dose-response	[8]
Weibull cumulative	Tunable shape parameter, recovers the monoexponential approach at $k = 1$	Three-parameter and identifiability is similar to Gompertz	[12]
Bi-exponential	Two-phase response (rapid plus slow)	Non-monotone rate of approach; estimation can be unstable	[10]
Spline / GAM mean structure	Non-parametric, captures relapse and treatment escape	Loses the parsimony argument; harder to calibrate to a single target trajectory	[13]
Latent change score / GMM	Allows participant-level trajectory heterogeneity in shape	Heavy machinery; identifiability requires more timepoints than a typical N-of-1 trial provides	[14]
Stochastic differential equation	Endogenous treatment-escape and relapse	Loses closed-form simulation; substantially heavier compute	[22]

8. Recommendations for pmsimstats-ng

1. **Document the Gompertz choice as a modeling assumption, not a mechanism.** The pmsimstats-ng manuscripts (notably the companion to this document at `analysis/report/01-dgp-mean-moderation-vs-mvn/` and the sensitivity papers at `analysis/report/02-carryover-sensitivity/` and `analysis/report/05-nof1-design-sensitivity/`) should state in the simulation-design section that the Gompertz is chosen for parsimony and tractability, that its sigmoidal shape is defensible but not the only

defensible choice, and that the power estimates are conditioned on this functional form.

2. **Add a Gompertz-versus-alternative sensitivity sweep.** A direct test of robustness would replace the Gompertz BR with a logistic 4PL and a Hill-form sigmoid at matched asymptote-and-inflection coordinates and report the resulting power changes. The infrastructure for such a sweep already exists in the `analysis/scripts/{carryover,nof1-design}-sensitivity/` pipelines and would require an additional axis in the cell tibble plus a small change to `simulation-core.R` to dispatch on response family. (*Note, 2026-10-07:* family dispatch now exists in the package, `trajectoryShape()` with the `br_family` argument of `generateData()`, and paper 07's driver `02-faithful-trajectory-sweep.R` (in `analysis/scripts/gompertz-evaluation` uses it; the Hill form is not among the implemented families.)

3. **Rename `modgompertz` or document the deviation from the Zwietering modified form.** Either the function name should be changed to reflect the actual modification (vertical-offset and rescale to origin) or the in-code roxygen documentation should make the deviation from Zwietering explicit. At present, a reader who recognizes 'modified Gompertz' from the bacterial-growth literature will encounter a different parameterization under the same name.

4. **Guard the rescale factor.** Add an explicit check on `(maxr - vert_offset)` for near-zero values and either issue a warning or fall back to the unmodified Gompertz at low `disp`.

5. **Consider participant-level variation in `maxr`.** A random-effects extension in which each simulated participant draws an asymptote $a_i \sim N(a_0, \tau_a^2)$ (code `maxr`) would accommodate plateau heterogeneity without abandoning the Gompertz family. This would also bring the framework into better alignment with the latent-class and finite-mixture formulations developed in the companion paper at `analysis/report/03-latent-class-mixture-application/`.

6. **Acknowledge response-then-relapse as a scope exclusion.** Until the framework supports non-monotone response trajectories, manuscripts should state explicitly that the simulation does not capture mid-treatment loss of benefit, and that the reported power estimates may be optimistic for clinical populations in which such loss is documented.

9. Summary

The Gompertz model is a defensible parametric template for the simulated symptom trajectories in pmsimstats-ng. It is parsimonious, numerically tractable, and has a long pedigree in adjacent biological domains. It is also a strong assumption: it imposes monotone non-decreasing trajectories with

a fixed population asymptote and a specific asymmetric inflection that may not match the empirical PTSD or chronic-pain symptom-trajectory literature. When this note was written (2026-05-07) the pmsimstats-ng manuscripts used the Gompertz without a sensitivity comparison against the established Emax / Hill alternative or against logistic and Weibull-cumulative families. (*Note, 2026-10-07:* paper 07 now supplies a comparison against the logistic, hyperbolic-tangent and piecewise-linear families at one design and $t_{1/2} = 0$; Emax / Hill and Weibull remain untested.) We recommend that future revisions either defend the Gompertz choice empirically against a candidate alternative or report a sensitivity sweep that demonstrates the power conclusions are robust to the choice of sigmoid family. The Gompertz alone should not bear the weight of a methodological recommendation about trial design.

References

Metadata were checked on 2026-10-07 against PubMed (entries 2 to 10, 15 to 21, 23, 24) and Crossref (entries 1, 11 to 13); entries 14 and 22 are books checked against the publisher record.

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